

Alert Logic Documentation

Campus Wastewater Health Alert Dashboard

Team: Python and Proteins

This document is the **Alert Logic** deliverable for the hackathon submission. It specifies, precisely and without ambiguity, how a raw wastewater reading becomes a Green / Yellow / Red alert, a 0–100 Health Score, a plain-language Reason, a Recommended Action, a Risk Trend, and a Confidence Score. Every threshold, formula, and example below is taken directly from the project's own `data_generator.py`, so this document and the running dashboard cannot drift out of sync.

1. Overview: the three-tier alert system

Every sampling reading -- one row per (site, day) -- is classified into exactly one of three levels:

Level	Meaning	Trigger condition
● Green	Within baseline / normal range	No marker breaches its Yellow or Red threshold
● Yellow	Early-warning: elevated, watch	At least one marker exceeds its Yellow threshold, and none exceed Red
● Red	Outbreak / contamination-level: act	At least one marker exceeds its Red threshold (checked first, independently of all others)

Design principle: Red is checked first and wins outright. A site with one marker deep in Red and everything else perfectly Green is still Red -- alert level is driven by the single worst marker, not an average, so a real localized problem can never be diluted into a false all-clear.

2. Markers monitored and their thresholds

Nine markers are tracked per reading. Eight have a documented Yellow/Red threshold; the ninth (Campylobacter) is deliberately left without one --

Marker	Unit	Yellow	Red	Notes
SARS-CoV-2	copies/100kL	50	150	Flow-normalized
Influenza A	copies/100kL	75	200	Flow-normalized
Norovirus	copies/100kL	50	150	Flow-normalized
E. coli	MPN/100mL	500	2500	Raw reading
Salmonella	MPN/100mL	100	500	Raw reading
Campylobacter	MPN/100mL	not set	not set	Tracked, no threshold yet
Turbidity	NTU	20	50	Raw reading
Antibiotic residues	ng/L	300	1000	Raw reading
pH	pH units	none (no Yellow band)	< 5.5 or > 9.0	Two-sided band

Rows shaded pink are the two documented exceptions to a standard single Yellow/Red cutoff -- explained just below.

2.1 pH: a two-sided band, not a single cutoff

pH is the one marker where 'too high' and 'too low' are both dangerous, so it doesn't fit a single ascending threshold. Instead it uses a safe band: **Red fires if pH < 5.5 or pH > 9.0**. No Yellow band is currently defined for pH -- a reading is either inside the safe band (Green, as far as pH is concerned) or outside it (Red). This is a known simplification, not an oversight: it's flagged in the Methodology tab as a candidate to refine with an intermediate Yellow band in a future iteration.

2.2 Campylobacter: tracked, but no threshold set

Campylobacter is measured and shown on every chart and export, but it currently has **no Yellow/Red threshold** and therefore **cannot trigger an alert on its own**. This was a deliberate decision, not a bug: rather than inventing a cutoff the team couldn't back with a source, the honest choice was to leave it unset and say so. It's excluded from Health Score, Confidence, and Reason calculations for exactly the same reason -- nothing downstream should imply a validated cutoff that doesn't actually exist.

3. Flow normalization for viral markers

Wastewater flow varies a lot -- by occupancy, weekday vs weekend, and weather (rain). A viral signal measured in a heavily-diluted, high-flow sample can look artificially low even if shedding hasn't changed, and vice-versa for a low-flow sample. To correct for this, SARS-CoV-2, Influenza A, and Norovirus are converted to a flow-normalized value before being compared against any threshold:

```
marker_norm = marker_raw * 100 / Flow_Rate_kL
```

i.e. "copies per 100 kL of flow" rather than a raw concentration. All three viral thresholds in the table above are defined on this normalized value, not the raw reading -- this is why the app reports both `Norovirus` (raw) and `Norovirus_norm` (the one alert logic actually uses).

4. Alert level decision algorithm

For every row, the algorithm evaluates two boolean conditions and picks the level in this fixed order:

```
RED = (SARS-CoV-2_norm > 150) OR (Influenza_A_norm > 200) OR
      (Norovirus_norm > 150) OR (E_Coli > 2500) OR
      (Salmonella > 500) OR (Turbidity_NTU > 50) OR
      (Antibiotic_Residues > 1000) OR (pH < 5.5 OR pH > 9.0)

YELLOW = (SARS-CoV-2_norm > 50) OR (Influenza_A_norm > 75) OR
         (Norovirus_norm > 50) OR (E_Coli > 500) OR
         (Salmonella > 100) OR (Turbidity_NTU > 20) OR
         (Antibiotic_Residues > 300)

Alert_Level = "Red" if RED
              "Yellow" elif YELLOW
              "Green" otherwise
```

Two properties are worth calling out explicitly, since they're easy to get wrong when re-implementing this logic elsewhere:

- **RED is evaluated in full before YELLOW is even considered.** A row is never Yellow because of one marker while quietly also breaching a Red threshold on another -- Red always wins.
- **Every condition is an OR across markers, not an AND.** A single marker crossing its line is sufficient; the other markers being perfectly normal does not soften the result.
- Campylobacter never appears in either condition.

5. Health Score (0–100)

Alert_Level gives three buckets; Health Score turns that into one continuous number, so a site trending worse is visible before it actually crosses into the next color:

```
score = 100
if Alert_Level == "Red": score -= 40
elif Alert_Level == "Yellow": score -= 15
if Turbidity_NTU > 15: score -= 5
score = clip(score, 0, 100)
```

In practice this yields four bands: **100** (clean Green), **~85** (Green but turbidity slightly elevated), **~60** (Yellow, or Red without a turbidity penalty), and **~55** (Red with the turbidity penalty applied) -- see the worked examples in section 9.

6. Reason: explainable, marker-by-marker

Rather than a bare "Red", every row gets a plain-language Reason listing exactly which marker(s) individually exceeded their own Yellow cutoff (or, for pH, fell outside the safe band). This is what "explainable AI" means in this dashboard -- it's a direct readout of which thresholds fired, not a black-box score:

- Each of the 7 thresholded markers (§2) contributes its own name if its value is above its Yellow cutoff -- e.g. "Norovirus", "E. coli", "Antibiotics".
- pH contributes "pH Anomaly" if it's outside the 5.5–9.0 band.
- Reported_Clinical_Cases contributes "Clinical Cases" if the day's count exceeds 5 -- the one reason component that isn't tied to a formal Yellow/Red threshold table, included because a clinical case spike is itself a meaningful corroborating signal.
- The matched names are joined with " + ", e.g. "Norovirus + Antibiotics + Clinical Cases".
- If nothing is elevated, Reason is simply "Normal Baseline".

7. Recommended action

Alert level	Recommended action
Red	Notify Health Centre, Increase sampling, Inspect sewage line
Yellow	Increase sampling frequency, Watch closely
Green	Routine monitoring

The recommendation is currently keyed only on Alert_Level (not on which specific marker triggered it). That's a simplification worth naming: a Red on pH and a Red on Norovirus get the same generic action list today, even though a real response would differ (a chemical-incident Red points toward a plumbing inspection; a pathogen Red points toward the Health Centre first). Flagged here as a natural next refinement rather than left implicit.

8. Risk Trend and Confidence Score

8.1 Risk Trend

Each site's Health Score is compared to its own value one day earlier (same site, previous calendar day):

```
delta = Health_Score(today) - Health_Score(yesterday)

Risk_Trend = "↑ Rising Health" if delta > 5
             "↓ Falling Health" if delta < -5
             "→ Stable" otherwise
```

Note the labels describe the *Health Score*'s direction, not "risk" in the colloquial sense -- "↑ Rising Health" means the site is getting *better* (score going up), and "↓ Falling Health" means it's getting *worse*. This is worth double-checking against the dashboard's own labeling so the two never say opposite things.

8.2 Confidence Score

Roughly 1.5% of sensor readings are randomly simulated as missing (sensor/lab dropouts). Confidence Score reports simply how much of that row's data actually arrived:

```
Confidence_Score = (# of the 10 sensor markers present, i.e. not NaN) / 10 × 100%
```

A Confidence Score below 100% is a signal to the user that an alert (or a clean Green) for that row is based on partial data, not a judgment that the reading itself is wrong.

9. Worked examples (pulled from the actual generated dataset)

These three rows are not illustrative made-up numbers -- they are exactly what running the current `generate_dataset()` produces.

9.1 Outbreak in progress — Beas Kund, 2026-10-11 (day 72 of the Norovirus bell curve)

Flow rate	139.5 kL
Norovirus (raw)	890.2 copies/mL
Norovirus (flow-normalized)	638.1 → above Red (150)
Reported clinical cases	29
Antibiotic residues	861.4 ng/L
Alert_Level	Red — Norovirus_norm alone exceeds its Red threshold
Health_Score	60 (100 – 40 for Red)
Reason	Norovirus + Antibiotics + Clinical Cases
Recommended action	Notify Health Centre, Increase sampling, Inspect sewage line
Risk_Trend	→ Stable
Confidence_Score	100%

9.2 Acute chemical incident — A1, 2026-10-19 (single-day pH crash)

pH	4.79 → below Red-low (5.5)
Turbidity	46.2 NTU → above Red (50) and above 15 (score penalty)
Alert_Level	Red — pH outside the safe band, independent of every other marker
Health_Score	55 (100 – 40 for Red – 5 turbidity penalty)
Reason	Turbidity + pH Anomaly
Recommended action	Notify Health Centre, Increase sampling, Inspect sewage line

9.3 Ordinary baseline day — Gulmohar Mess, 2026-08-10

Alert_Level	Green — no marker breaches its Yellow or Red threshold
Health_Score	100 (no penalties)
Reason	Normal Baseline

Recommended action	Routine monitoring
Confidence_Score	90% (one or more sensor readings were missing that day)

10. Known limitations, stated honestly

- **Campylobacter has no threshold** and cannot trigger an alert (2.2) -- a candidate for a future threshold once a source can be cited.
- **pH has no Yellow band**, only the two Red cutoffs (§2.1) -- an intermediate warning band is a natural next step.
- **Recommended action is keyed only on Alert_Level**, not on which marker triggered it (§7) -- today's Red response text doesn't yet differentiate a pathogen incident from a chemical one.
- **Threshold sourcing**: per the project README, the Yellow/Red cutoffs are grounded in CPCB water quality standards and published wastewater-based-epidemiology literature; the exact citation for each number is intended to be filled into the Methodology tab's Source column rather than asserted here.
- **Risk Trend compares only to the immediately preceding day** (a 1-day delta), so a single noisy reading can flip the label; it is a short-term indicator, not a smoothed multi-day trend.